



Hybrid Analytical–Machine-Learning Surrogate Modeling for Buckling Prediction and Design of Functionally Graded Porous Nanobeams

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Abstract: Accurate buckling prediction of functionally graded porous (FGP) nanobeams requires the simultaneous consideration of material gradation, porosity distribution, transverse shear deformation, and small-scale effects. Conventional analytical models provide a sound mechanical basis; however, they become less convenient when repeated evaluations and multi-objective design decisions are required. This study develops a hybrid analytical-machine-learning (ML) framework for predicting and interpreting the buckling behavior of FGP nanobeams. An analytical model was formulated using exponential shear deformation theory and Eringen’s nonlocal elasticity theory. The governing equations were derived from the minimum total potential energy principle and solved by Navier’s method for simply supported boundary conditions. The formulation was verified against published benchmark results and was subsequently used to generate 540 configurations covering different slenderness ratios, nonlocal parameters, porosity coefficients, and porosity distributions. A residual-learning surrogate was then constructed by combining a regularized Ridge baseline with nonlinear ML models. Nested cross-validation (CV) showed that the hybrid support vector regression model with a radial basis function produced the lowest prediction errors, with a root mean square error (RMSE) of 0.00729, a mean absolute error (MAE) of 0.00309, and an R^2 of 0.999900. The analytical and surrogate results showed that the buckling load decreases with increasing slenderness ratio, nonlocal parameter, and porosity coefficient, while the functionally graded X-pattern (FGX) distribution retains the highest buckling resistance. The conformal prediction intervals achieved 97.04% empirical coverage. SHapley Additive exPlanations (SHAP) analysis identified the slenderness ratio as the dominant input, and Nondominated Sorting Genetic Algorithm II (NSGA-II) yielded a Pareto-based compromise between porosity and buckling resistance. The proposed framework demonstrates how analytical mechanics, residual learning, uncertainty quantification, model interpretation, and multi-objective optimization can be coordinated within a unified engineering design procedure.

Keywords: Functionally graded porous nanobeam; Buckling prediction; Exponential shear deformation theory; Nonlocal elasticity; Residual learning; Surrogate modeling; Conformal prediction; Multi-objective optimization

1 Introduction

Functionally graded porous (FGP) structures combine low weight with high strength, energy absorption capacity, and favorable thermal and acoustic insulation. Their mechanical and functional properties can be tailored by controlling the size, density, gradation, and spatial distribution of pores [1–3]. These characteristics have supported the use of FGP structures in aerospace, automotive, civil, and biomedical engineering. Their increasing engineering relevance has also led to sustained interest in modeling their mechanical behavior under different loading and operating conditions. In particular, the buckling response of FGP beams has been examined using a range of structural theories and solution procedures. Magnucki and Stasiewicz [4] studied the buckling of porous beams under compressive loading using a broken-line hypothesis and the total potential energy principle. Magnucka-Blandzi and Magnucki [5] investigated the bending and buckling behavior of simply supported sandwich beams with metal foam cores and considered the associated design problem through two different approaches. Chen et al. [6] analyzed the bending

and buckling of shear-deformable FGP beams using Timoshenko beam theory and the Ritz method, showing that symmetric porosity distributions produced higher buckling loads and smaller transverse deflections than asymmetric distributions. Chen et al. [7] subsequently examined the nonlinear vibration and post-buckling behavior of metal-foam FGP beams reinforced with graphene platelets. Kitipornchai et al. [8] investigated the free vibration and elastic buckling of multilayered functionally graded (FG) metal-foam beams reinforced with graphene platelets. Nguyen et al. [9] developed solutions for the bending, buckling, and free vibration of shear-deformable FGP beams using Lagrange's principle and the Ritz method. Wu et al. [10] compared classical and shear-deformation beam theories for the buckling and free-vibration analysis of FGP beams through the generalized differential quadrature method.

Advances in nanotechnology have enabled the development of nanoelectromechanical systems (NEMS) with distinctive mechanical, electrical, and thermal characteristics. These systems are used in medical, optical, electronic, and energy-related applications and commonly contain nanoscale structural components such as beams, plates, and shells [11]. Nanobeams serve as one-dimensional load-carrying or sensing components in devices such as atomic-force-microscope cantilevers, nanoactuators, resonators, and nanosensors. The performance and accuracy of such devices therefore depend strongly on the mechanical response of their nanobeam components [12–16].

At the nanoscale, structural behavior may differ substantially from that predicted for macroscale components because conventional continuum theories do not account for size-dependent effects. Nonlocal strain-gradient theory, modified couple-stress theory, and nonlocal elasticity theory have consequently been developed or adopted to represent these effects. Among these formulations, Eringen's nonlocal elasticity theory remains widely used in nanobeam analysis because it introduces the small-scale effect through a relatively concise constitutive framework [17–21]. The mechanical behavior of nanobeams has been investigated through several continuum formulations and numerical solution techniques [22–24]. Reddy [25] derived nonlocal governing equations for several beam theories and used a finite-element formulation to study the bending, buckling, and free vibration of nanobeams. Aydogdu [26] developed a generalized nonlocal beam model for their static and dynamic analysis. Zhang et al. [24] proposed a two-parameter hybrid nonlocal beam model to represent the static and dynamic responses of micro- and nanobeams. Thai [27] and Thai and Vo [28] combined nonlocal elasticity with different shear-deformation theories to investigate nanobeam bending, buckling, and free vibration. Ebrahimi and Salari [29] examined the free vibration and thermal buckling of FG Timoshenko nanobeams using a Navier solution. Civalek and Demir [30] modeled microtubules as nonlocal Euler–Bernoulli beams embedded in an elastic medium and evaluated their buckling response using the finite-element method. Eltaher et al. [31] also employed a finite-element formulation to study the bending and vibration of FG Euler–Bernoulli nanobeams. Aria and Friswell [32] developed a size-dependent Timoshenko beam model within the nonlocal elasticity framework for the finite-element analysis of buckling and free vibration in FG nanobeams.

Reliable prediction of static buckling in FGP nanobeams is important for the design of stable NEMS components because their response is governed jointly by stiffness gradation, porosity distribution, beam geometry, and size-dependent behavior [8, 31]. Shafiei and Kazemi [33] investigated the buckling of two-dimensional tapered FGP nano- and microbeams by combining nonlocal elasticity with modified couple-stress theory. Mirjavadi et al. [34] used the generalized differential quadrature method to analyze the buckling and nonlinear vibration of FGP nanobeams. Aria et al. [35] studied the free vibration and buckling responses of FGP nanobeams subjected to thermal loading. More recently, Madasamy and Wang [36] examined the buckling behavior of bidirectional FGP nanobeams under moving loads in hygrothermal environments using a higher-order Haar wavelet method together with the generalized differential quadrature method. These studies have established useful analytical and computational foundations, but evaluating numerous combinations of geometric, material, and nonlocal parameters remains demanding when the analysis is extended to repeated prediction, uncertainty assessment, and design optimization.

Machine-learning (ML) methods provide an alternative means of constructing surrogate representations of structural responses across multidimensional parameter spaces [37]. Their use in the modeling of micro- and nanoscale structures has increased in recent years, particularly where repeated evaluations are required [38–40]. Tran et al. [41] combined an artificial neural network (ANN) with an optimization procedure to predict the buckling and vibration responses of FGP microplates. Tariq et al. [42] applied boosting-based ensemble models to investigate the effects of multiple parameters on the vibration response of porous nanobeams and used SHapley Additive exPlanations (SHAP) to assess the contribution of individual inputs. Al-Dahidi and Sari [43] compared regression models, ANNs, and support vector machines for predicting the buckling response of FGP nanobeams under different boundary conditions. Cheng et al. [44] used an ANN to estimate the effects of eight parameters on the vibration behavior of FGP nanobeams. Although these studies demonstrate the predictive capacity of ML, most treat the learning model primarily as a direct input–output approximation. Less attention has been paid to the structured integration of an analytical mechanics formulation, residual learning, validation without information leakage, uncertainty quantification, interpretable parameter assessment, and multi-objective engineering design within a single framework.

The present study addresses this methodological gap by developing a hybrid analytical–ML surrogate framework

for the buckling prediction and design of FGP nanobeams. The analytical component was formulated using exponential shear deformation theory, with three porosity distributions defined according to the Gibson–Ashby model. The exponential shear function satisfies the traction-free conditions at the upper and lower surfaces, removing the need for a shear correction factor. Eringen’s nonlocal elasticity theory was incorporated to represent small-scale effects, and the governing equations were derived from the minimum total potential energy principle. Navier’s method was then used to obtain the buckling loads of simply supported FGP nanobeams, and the formulation was verified against published benchmark results. Parametric analyses were conducted to determine the effects of the nonlocal parameter, slenderness ratio, porosity coefficient, and porosity distribution on the dimensionless buckling load (DBL). The resulting analytical dataset was subsequently used to construct a residual-learning surrogate in which a regularized mechanics-informed baseline represented the principal response trend, and nonlinear ML models captured the remaining discrepancy. The integrated framework further combined nested cross-validation (CV), conformal uncertainty quantification, SHAP-based interpretation, and Nondominated Sorting Genetic Algorithm II (NSGA-II) multi-objective optimization. This arrangement links analytical mechanics, predictive modeling, uncertainty assessment, model interpretation, and design exploration while preserving the physical trends of the underlying buckling formulation.

2 Formulation of the Problem

2.1 Functionally Graded Porous Nanobeam

The FGP nanobeam was considered with length (L), thickness (h), and width (b), which were located on the x , y , and z axes, respectively. The beam was subjected to an axial load N_0 as shown in Figure 1a. The FGP nanobeam’s material properties were assumed to either change or remain constant along the thickness direction, depending on the specific porosity distribution types (PDTs). Specifically, the functionally graded X-pattern (FGX) and functionally graded O-pattern (FGO) configurations denote a continuous decrease or increase in porosity from the mid-plane of the nanobeam to its top and bottom surfaces, respectively. Uniform distribution (UD) indicates that the porosity is distributed uniformly [10]. The effective Young’s and shear moduli can be defined as follows, respectively:

$$\begin{aligned} \begin{bmatrix} E(z) \\ G(z) \end{bmatrix} &= \begin{bmatrix} E_{\max} \\ G_{\max} \end{bmatrix} [1 - \alpha_0 \eta(z)], \\ G_{\max} &= \frac{E_{\max}}{2(1 + \nu)} \end{aligned} \quad (1)$$

where, ν , $E_{\max}$, and $G_{\max}$, represent the Possion’s ratio, the maximum values of the Young’s and shear moduli of the FGP nanobeam, respectively, as well as $\eta(z)$ and α_0 represent the porosity distribution function and porosity coefficient, respectively, and are defined as follow:

$$\eta(z) = \begin{cases} \cos\left(\frac{\pi z}{h}\right) & \text{(FGX)} \\ \cos\left(\left|\frac{\pi z}{h}\right| - \frac{\pi}{2}\right) & \text{(FGO)} \\ \eta_0 & \text{(UD)} \end{cases} \quad (2)$$

$$\alpha_0 = 1 - \frac{E_{\min}}{E_{\max}} = 1 - \frac{G_{\min}}{G_{\max}} \quad (3)$$

where, η_0 is the PDT function for the UD, $E_{\min}$ and $G_{\min}$ are the minimum values of Young’s and shear moduli of the layer with the highest porosity, respectively.

According to the Gibson-Ashby micromechanical model, the relation between Young’s modulus and mass density is defined as [45]:

$$\frac{E(z)}{E_{\max}} = \left(\frac{\rho(z)}{\rho_{\max}} \right)^2 \quad (4)$$

where, $\rho_{\max}$ is the mass density of non-porous material. The effective mass density of the FGP nanobeam can be formulated by substituting Eq. (4) into Eq. (1),

$$\rho(z) = \rho_{\max} \sqrt{1 - \alpha_0 \eta(z)} \quad (5)$$

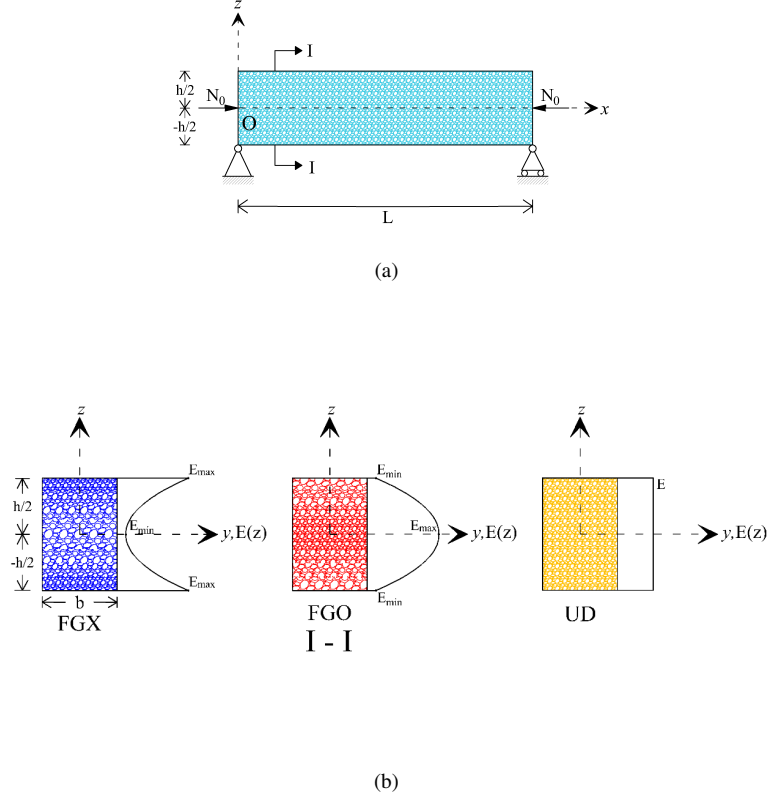


Figure 1. (a) The geometry of the functionally graded porous nanobeam; (b) the variation of the porosity distribution and Young's moduli through the thickness direction

Note: FGX — functionally graded X-pattern; FGO — functionally graded O-pattern; UD — uniform distribution.

FGP nanobeam with distinct PDTs must have equal total mass, thus η_0 can be acquired:

$$\int_{-h/2}^{h/2} \rho_{\max} \sqrt{1 - \alpha_0 \eta(z)} dz = \int_{-h/2}^{h/2} \rho_{\max} \sqrt{1 - \alpha_0 \eta_0} dz \quad (6)$$

$$\eta_0 = \frac{1}{\alpha_0} \left(1 - \left[\frac{1}{h} \int_{-h/2}^{h/2} \sqrt{1 - \alpha_0 \eta(z)} dz \right]^2 \right) \quad (7)$$

The porosity coefficient for mass density α_m can be defined as:

$$\alpha_m = 1 - \frac{\rho_{\min}}{\rho_{\max}}, 0 < \alpha_m < 1 \quad (8)$$

where, $\rho_{\max}$ and $\rho_{\min}$ are the maximum and the minimum values of mass density, respectively.

2.2 Kinematics

Based on the present theory, axial displacement u and transverse displacement w of any point of the FGP nanobeam can be given as:

$$\begin{aligned} u(x, z) &= u_0(x) - z \frac{\partial w_0}{\partial x} + f(z) \varphi(x) \\ w(x) &= w_0(x) \\ f(z) &= z e^{-2(z/h)^2} x \end{aligned} \quad (9)$$

where, u_0 and w_0 are axial and transverse displacements of any point on the midplane of the FGP nanobeam, φ is shear slope. $f(z)$ is the exponential shear-strain function which satisfies the traction-free boundary conditions on the nanobeam's outer surfaces [46]. The strain components associated with the displacement field are:

$$\varepsilon_x = \frac{\partial u_0}{\partial x} - z \frac{\partial^2 w_0}{\partial x^2} + f(z) \frac{\partial \varphi}{\partial x} \quad (10)$$

$$\begin{aligned} \gamma_{xz} &= g(z) \varphi \\ g(z) &= f'(z) \end{aligned} \quad (11)$$

where ε_x , γ_{xz} and $g(z)$ are the axial strain, transverse shear strain and the first derivative of the shear-strain function, respectively.

The normal σ_x and transverse τ_{xz} shear stresses at any point in a nanobeam can be obtained using Hooke's Law [3]:

$$\begin{aligned} \sigma_x &= E(z) \varepsilon_x \\ \tau_{xz} &= G(z) \gamma_{xz} \end{aligned} \quad (12)$$

2.3 Governing Equations

According to the principle of minimum total potential energy, the first variation of the total potential energy of the system must be zero. Thus, governing equations of the FGP nanobeam can be derived as follows [47]:

$$\delta \Pi = \delta U - \delta V = 0 \quad (13)$$

where, δU and δV are the first variation of the strain energy and the work done by external forces, respectively.

The first variation of the strain energy of the FGP nanobeam can be stated as:

$$\begin{aligned} \delta U &= b \int_0^L \int_{-h/2}^{h/2} (\sigma_x \delta \varepsilon_x + \tau_{xz} \delta \gamma_{xz}) dz dx \\ &= \int_0^L \left(N \frac{\partial \delta u_0}{\partial x} - M^b \frac{\partial^2 \delta w_0}{\partial x^2} - M^s \frac{\partial \delta \varphi}{\partial x} + Q \delta \varphi \right) dx \end{aligned} \quad (14)$$

where, N , M^b , M^s and Q are force and moment resultants which can be obtained by integrating stress components over the cross-sectional area as follows:

$$\begin{aligned} \begin{bmatrix} N \\ M^b \\ M^s \end{bmatrix} &= b \int_{-h/2}^{h/2} \begin{bmatrix} 1 \\ z \\ f(z) \end{bmatrix} \sigma_x dz \\ Q &= b \int_{-h/2}^{h/2} g(z) \tau_{xz} dz \\ N &= A_{11} \frac{\partial u_0}{\partial x} - B_{11} \frac{\partial^2 w_0}{\partial x^2} + C_{11}^s \frac{\partial \varphi}{\partial x} \\ M^b &= B_{11} \frac{\partial u_0}{\partial x} - D_{11} \frac{\partial^2 w_0}{\partial x^2} + F_{11}^s \frac{\partial \varphi}{\partial x} \\ M^s &= C_{11}^s \frac{\partial u_0}{\partial x} - F_{11}^s \frac{\partial^2 w_0}{\partial x^2} + H_{11}^s \frac{\partial \varphi}{\partial x} \\ Q &= J_{11}^s \varphi \end{aligned} \quad (15)$$

where, A_{11} , B_{11} , C_{11}^s , D_{11} , F_{11}^s , H_{11}^s and J_{11}^s are stiffness coefficients and are expressed as:

$$\begin{aligned} (A_{11}, B_{11}, D_{11}) &= b \int_{-h/2}^{h/2} E(z) [1, z, z^2] dz \\ (C_{11}^s, F_{11}^s, H_{11}^s) &= b \int_{-h/2}^{h/2} E(z) [f(z), z f(z), f^2(z)] dz \\ J_{11}^s &= b \int_{-h/2}^{h/2} G(z) g^2(z) dz \end{aligned} \quad (16)$$

The first variation of work done by axial compressive load N_0 can be expressed as:

$$\delta V = \int_0^L N_0 \frac{\partial w}{\partial x} \frac{\partial \delta w}{\partial x} dx \quad (17)$$

The governing equations can be derived by introducing Eq. (14) and Eq. (17) into Eq. (13), consequently applying integration by parts, then setting the coefficients δu_0 , δw_0 and $\delta \varphi$ to zero:

$$\begin{aligned} \delta u_0 : \frac{\partial N}{\partial x} &= 0 \\ \delta w_0 : \frac{\partial^2 M^b}{\partial x^2} &= N_0 \frac{\partial^2 w_0}{\partial x^2} \\ \delta \varphi : \frac{\partial M^s}{\partial x} - Q &= 0 \end{aligned} \quad (18)$$

2.4 Eringen's Nonlocal Elasticity

The nonlocal stress state of a specific reference point in a continuum is a function of not only the strain at that exact point but also the strain field throughout the entire body, according to Eringen's nonlocal elasticity theory. The nonlocal stresses can be derived using Eringen's differential form [17, 18]:

$$\left(1 - \mu \frac{\partial^2}{\partial x^2}\right) \begin{bmatrix} \sigma_x \\ \tau_{xz} \end{bmatrix} = \begin{bmatrix} E(z) & 0 \\ 0 & G(z) \end{bmatrix} \begin{bmatrix} \varepsilon_x \\ \gamma_{xz} \end{bmatrix} \quad (19)$$

where, $\mu = (e_0 a)^2$ is the nonlocal parameter, which is responsible for capturing small-scale effects. α and e_0 are the internal characteristic length and a material constant, respectively. Literature studies suggest that $e_0 a \leq 2$ nm for single-walled carbon nanotubes [48]. Note that, when $\mu = 0$ is considered, the small-scale effect vanishes and the formulation reduces to the classical (local) beam.

The stress resultants are modified by introducing Eq. (10) and (11) into Eq. (19) and subsequent results into Eq. (15):

$$\begin{aligned} \left(1 - \mu \frac{\partial^2}{\partial x^2}\right) N &= A_{11} \frac{\partial u_0}{\partial x} - B_{11} \frac{\partial^2 w_0}{\partial x^2} + C_{11}^s \frac{\partial \varphi}{\partial x} \\ \left(1 - \mu \frac{\partial^2}{\partial x^2}\right) M^b &= B_{11} \frac{\partial u_0}{\partial x} - D_{11} \frac{\partial^2 w_0}{\partial x^2} + F_{11}^s \frac{\partial \varphi}{\partial x} \\ \left(1 - \mu \frac{\partial^2}{\partial x^2}\right) M^s &= C_{11}^s \frac{\partial u_0}{\partial x} - F_{11}^s \frac{\partial^2 w_0}{\partial x^2} + H_{11}^s \frac{\partial \varphi}{\partial x} \\ \left(1 - \mu \frac{\partial^2}{\partial x^2}\right) Q &= J_{11}^s \varphi \end{aligned} \quad (20)$$

By replacing Eq. (20) into Eq. (18), the nonlocal governing equations can be expressed in terms of unknown displacements are obtained:

$$\begin{aligned} \delta u_0 : A_{11} \frac{\partial^2 u_0}{\partial x^2} - B_{11} \frac{\partial^3 w_0}{\partial x^3} + C_{11}^s \frac{\partial^2 \varphi}{\partial x^2} &= 0 \\ \delta w_0 : B_{11} \frac{\partial^3 u_0}{\partial x^3} - D_{11} \frac{\partial^4 w_0}{\partial x^4} + F_{11}^s \frac{\partial^3 \varphi}{\partial x^3} &= N_0 \left(\frac{\partial^2 w_0}{\partial x^2} - \mu \frac{\partial^4 w_0}{\partial x^4} \right) \\ \delta \varphi : C_{11}^s \frac{\partial^2 u_0}{\partial x^2} - F_{11}^s \frac{\partial^3 w_0}{\partial x^3} + H_{11}^s \frac{\partial^2 \varphi}{\partial x^2} - J_{11}^s \varphi &= 0 \end{aligned} \quad (21)$$

3 Navier Solution

This section presents the Navier solution method to analytically solve the buckling problem of simply supported FGP nanobeams. The simply supported boundary conditions are satisfied as follows:

$$(u_0, w_0, M^b, M^s) \Big|_{x=0}^{x=L} = 0 \quad (22)$$

The displacement components are expressed as trigonometric series, containing unknown coefficients to satisfy the boundary conditions [49]:

$$\begin{Bmatrix} u_0 \\ w_0 \\ \varphi \end{Bmatrix} = \sum_{n=1}^{\infty} \begin{Bmatrix} U_n \cos(\beta x) \\ W_n \sin(\beta x) \\ \Phi_n \cos(\beta x) \end{Bmatrix} \quad (23)$$

where, U_n , W_n , and Φ_n are the unknown coefficients of displacement. $\beta = n\pi/L$ and n is the mode number.

The nontrivial solution of the buckling of FGP nanobeam is obtained by substituting Eq. (23) into Eq. (21):

$$\begin{bmatrix} A_{11}\beta^2 & -B_{11}\beta^3 & C_{11}^s\beta^2 \\ -B_{11}\beta^3 & D_{11}\beta^4 - N_0(\beta^2 + \mu\beta^4) & -F_{11}^s\beta^3 \\ C_{11}^s\beta^2 & -F_{11}^s\beta^3 & H_{11}^s\beta^2 + J_{11}^s \end{bmatrix} \begin{Bmatrix} U_n \\ W_n \\ \Phi_n \end{Bmatrix} = 0 \quad (24)$$

4 Hybrid Machine Learning Approach

The objective of the current proposed hybrid ML approach was to transform the verified solution of the governing formulation of FGP nanobeam to a rapid, interpretable surrogate model to assist engineering decisions based on the powerful prediction performance of the hybrid ML approach. In addition, to explore the ability of the hybrid ML approach in predicting complex structural behavior. The main concept of the proposed approach aligns with recent ML implementations in predicting buckling performance of FGP beams and other thin-walled structures [50–52]. The novelty of the proposed hybrid ML modeling is combining analytical features, baseline with residual learning, preventing leakage, uncertainty analysis, and explanatory artificial intelligence by SHAP, in addition to multi-objective design within one unified framework. As seen in Figure 2, the data of 540 unique samples were preprocessed, including their L/h , PDT, α_0 , μ , and $\bar{N}$ values. The Python-PyCharm code included standardization and checking the alignment of each sample with its accumulated values, and implemented a one-hot encoder to handle the PDT (UD, FGO, and FGX) as numeric values. These features were the main and basic features used to map the learning approach, then further features were integrated to expose the interaction trends between features and the dominant geometry. These features were generated by creating relations between L/h , α_0 , and μ . However, to minimize the large variation between L/h values, a logarithmic space was used to learn the positive target. However, to avoid using any information from the outer validation fold, all preprocessing procedures were accomplished inside the CV loop. Based on the literature, since ensemble boost models, ANN, and support vector revealed significant prediction performance, the current hybrid ML framework considers implementing all of these models in addition to Gaussian process regression (GPR). The ensemble models included eXtreme Gradient Boosting (XGBoost), histogram gradient boosting (HistGB), random forest (RF), and extra trees. For ANN, the multilayer perceptron (MLP) was used. The support vector regression was implemented based on radial basis (support vector regression-radial basis function, SVR-RBF). However, all algorithms were evaluated two times; first as a direct regression model, then as hybrid ML modeling. Some of the implemented parameters for ML algorithms and their value are illustrated in Table 1. However, the code was prepared and processed based on selecting the best combination of parameters that governed the optimum results.

The hybrid scheme has combined the nonlinear residual-learning model with the regularized Ridge model. Initially, the Ridge regression has estimated the principal trend of the DBL (on a logarithmic scale). Subsequently, the residual was defined as:

$$r_i = t_i - \hat{t}_{\text{base}}(x_i) \quad (25)$$

where, t_i and $\hat{t}_{\text{base}}$ stand for the logarithmic of the analytical DBL and the prediction of Ridge, respectively. while x_i is equivalent to x deified in Figure 2b.

However, by using the training data only, the parameter of Ridge regularization was selected as:

$$\lambda \in \{0.01, 0.1, 1, 10\} \quad (26)$$

Following that, the r_i , was set as the target to be predicted by the nonlinear model, causing the final hybrid prediction to be redefined as:

$$\hat{N}_{hyb}(x) = \exp [\hat{r}_{\text{ML}}(x) + \hat{t}_{\text{base}}(x)] \quad (27)$$

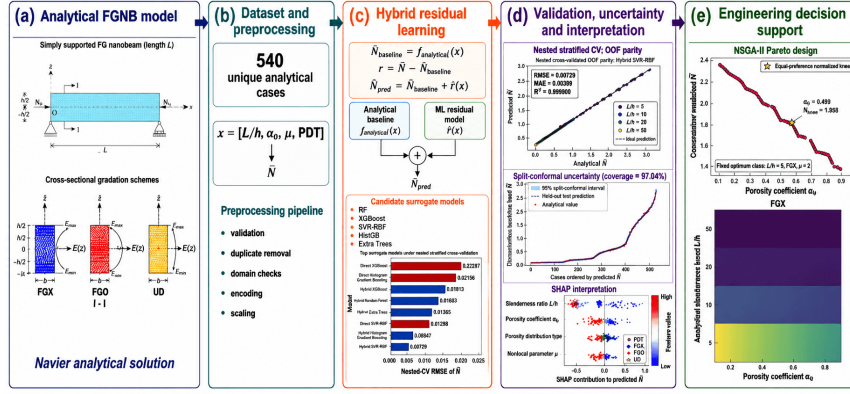


Figure 2. Hybrid-analytical-machine learning (ML) modeling workflow: (a) analytical modeling for functionally graded (FGP) nanobeam with various porosity distribution types (PDTs); (b) collecting the dataset and preprocessing within the validation loop; (c) the implemented hybrid residual learning and finding the best candidate model; (d) applying uncertainty and validation check and (e) applying optimization for assisting engineering decision

Note: FG — functionally graded; FGX — functionally graded X-pattern; FGO — functionally graded O-pattern; UD — uniform distribution; RF — random forest; XGBoost — eXtreme Gradient Boosting; SVR-RBF — support vector regression-radial basis function; HistGB — histogram-based gradient boosting; CV — cross-validation; RMSE — root mean square error; OOF — out-of-fold; MAE — mean absolute error; SHAP — SHapley Additive exPlanations.

Table 1. Machine learning (ML) algorithms and their parameters

ML Algorithm	Parameter	Parameter Value
RF	n_estimators	300–600
	Extra trees	300–600
XGBoost	n_estimators	300–900
	Learning rate	0.20–0.10
HistGB	Regularization	0.1–10
	Learning rate	0.03–0.01
SVR-RBF	Regularization	0–1
	Leaf nodes	15–63
GPR	Kernel coefficient	Scale: 0.01, 0.05, 0.10
	Penalty	1, 10, 50, 200
ANN-MLP	Matérn kernels	3/2, 5/2
	Stability term	10^{-8} , 10^{-6} , 10^{-4}
	Hidden layers	(64,32), (96,48), (128,64,32)
	Solver	Adam
	Activation function	ReLU

Note: RF — random forest; XGBoost — eXtreme Gradient Boosting; HistGB — histogram-based gradient boosting; SVR-RBF — support vector regression-radial basis function; GPR — gaussian process regression; ANN-MLP — artificial neural network - multi-layer perceptron; ReLU — Rectified Linear Unit.

where, $\hat{r}_{ML}(x)$ defined the predicted residual by ML, while the exponential was used to turn back the converted logarithmic scale to its original scale. Accordingly, Ridge represented the trend of principal response, while the ML corrected the remaining nonlinear discrepancy. However, the best selected residual ML learner model was revealed to be SSVR-RBF, as demonstrated in Section 5. The expression of SVR-RBF was integrated through the LIBSVM library in PyCharm, which used the well-known formulation of SVR network [53]. Nested CV was also used in selecting the model and hyperparameters. The architecture was built by considering five comprised and shuffled folds as an outer layer, which were joined by L/h -PDT using 42 random seeds. Each single outer has contained an inner 3-fold search, which created six hyperparameter sampled combination per each model. Consequently, each case has received one-out-of-fold (OOF) prediction from an outer unseen model (untrained model). The OOF was selected because it offers comprehensive evaluation across the entire sample size while preserving validation integrity. Furthermore, the stability of the model was evaluated by repeating 5-fold stratified validation in 10 repetitions, where the leave-one-level-out and learning curve were tested for each μ , L/h and α_0 , considering three error metrics as

R-squared (R^2), mean square error (MAE), and root mean square error (RMSE) to measure accuracy. The predicted results were compared with analytical findings, while the uncertainty was implemented using the disjoint split-conformal process, using separate folds for calibration, training, and testing [54]. Furthermore, the SHAP was used to inspect the importance of each feature, while the PDT was combined with SHAP and displayed different categories in different colors. Lastly, as a multi-objective design optimization, the validated hybrid SVR-RBF surrogate model was coupled with NSGA-II. The optimization approach continuously elevated the conservative buckling load α_0 to the maximum while considering $\mu = 2$, since the range of μ was (0, 1, 2, 3, and 4), 2 was selected because it presents the center intermediate value. Besides, to avoid unsupported extrapolation, optimization variables were limited to the analytical domain only. However, because the buckling resistance reduced with increasing porosity, which was concluded by testing several nanobeam designs and recording their porosity, a single design was not able to offer the maximum value of both. Due to this, the Pareto front was used to determine the optimum available candidate to select between larger porosity and greater buckling resistance [55]. The Pareto front was developed based on NSGA-II and the prediction of the trained hybrid model.

5 Numerical Results and Discussion

5.1 Verification Study

In this section, the numerical results are compared with previous results to verify current formulation. Note that, for all studies, $n = 1$ was considered.

In Table 2, the DBL values of FGP beams ($\mu = 0$) against α_0 for different PDTs were compared with the results of Wu et al. [10]. Here, the following material and geometric properties were assumed as: $E = 200$ GPa, $\nu = 0.3$ and $h = 1$ m, slenderness ratio $L/h = 10$. For the DBL, the following formulation was used, unless stated otherwise:

$$\bar{N} = \frac{100N_0}{Eh} \quad (28)$$

Table 2. The comparison of dimensionless buckling loads (DBLs) of functionally graded porous (FGP) beams versus α_0

α_0	$\bar{N}$					
	Wu et al. [10]			Present		
	FGX	FGO	UD	FGX	FGO	UD
0.00	0.8019	0.8019	0.8019	0.8020	0.8020	0.8019
0.90	0.5210	0.1667	0.3049	0.5185	0.1668	0.5210
0.99	0.4856	0.1022	0.2349	0.4799	0.1023	0.4856

Note: FGX — functionally graded X-pattern; FGO — functionally graded O-pattern; UD — uniform distribution.

In Table 3, DBLs of isotropic nanobeams ($\alpha_0 = 0$) versus L/h and μ with the results of Thai and Vo [28] were compared. Furthermore, following formulation was used for DBL:

$$\bar{N} = \frac{N_0 L^2}{EI} \quad (29)$$

Table 3. The comparison of dimensionless buckling loads (DBLs) of isotropic nanobeams versus L/h and μ

Reference	L/h	$\bar{N}$				
		μ (nm ²)				
		0	1	2	3	4
Present	5	8.9573	8.1527	7.4807	6.9111	6.4220
	10	9.6242	8.7597	8.0377	7.4256	6.9002
	20	9.8071	8.9261	8.1904	7.5667	7.0312
	100	9.8671	8.9807	8.2405	7.6130	7.0743
[28]	5	8.9533	8.1490	7.4773	6.9079	6.4191
	10	9.6231	8.7587	8.0367	7.4247	6.8994
	20	9.8068	8.9258	8.1901	7.5665	7.0310
	100	9.8671	8.9807	8.2405	7.6130	7.0743

It can be seen from Table 2 and Table 3, that the obtained results from present theory were in good agreement with previous studies.

5.2 Parametrical Studies

5.2.1 Analytical solution

The conducted parametric studies to elucidate the effect of nonlocal parameter, slenderness ratio L/h , PDT and α_0 on DBLs of the FGP nanobeams is addressed in this section. The following material and geometrical properties were assumed: $E = 200$ GPa, $\nu = 0.3$, $L = 10$ nm and $b = 1$ nm, the first mode number of buckling was considered ($n = 1$) and for the DBL, the following formulation was used:

$$\bar{N} = \frac{100N_0}{Eh} \quad (30)$$

Figure 3 visualizes the variation of DBL values of FGP nanobeam versus μ for different PDT and α_0 , and $L/h = 10$. It can be seen from Figure 2, DBLs decrease as μ and α_0 values increase for all PDTs.

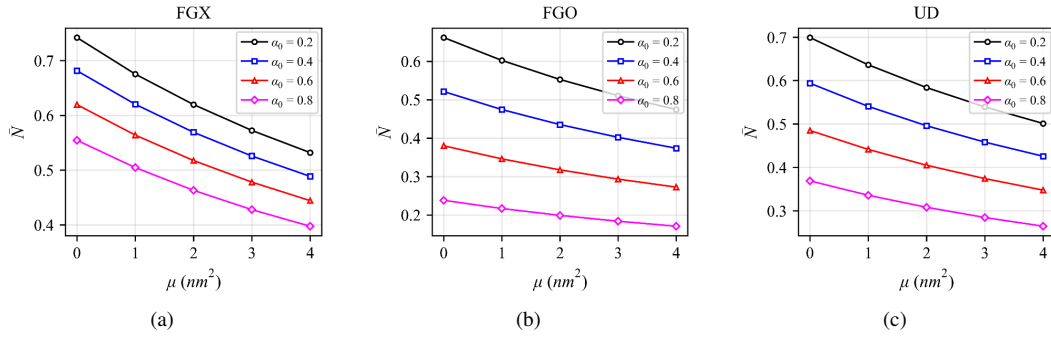


Figure 3. The variation of dimensionless buckling loads (DBLs) of functionally graded porous (FGP) nanobeam versus μ : (a) functionally graded X-pattern (FGX); (b) functionally graded O-pattern (FGO); (c) uniform distribution (UD)

In Figure 4, the variation of DBLs versus L/h for different α_0 is presented for different PDTs and $\mu = 2\text{nm}^2$. Besides, it is revealed that an increase in L/h is not only leads to reduction in DBLs but also causes DBL values from distinct α_0 converge.

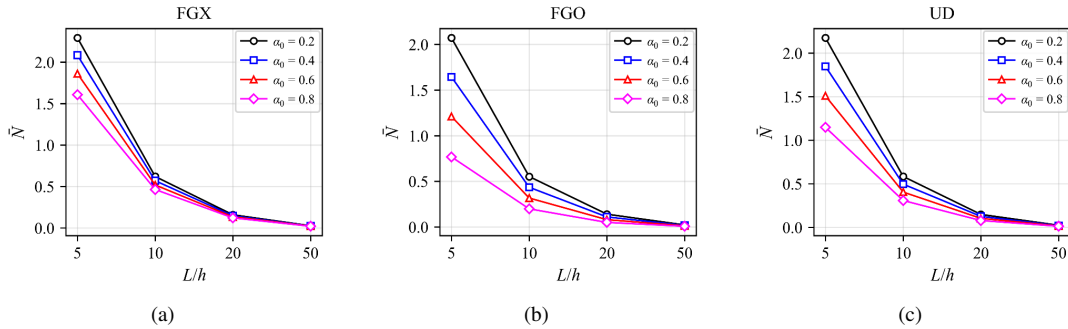


Figure 4. The variation of dimensionless buckling loads (DBLs) of functionally graded porous (FGP) nanobeam versus L/h : (a) functionally graded X-pattern (FGX); (b) functionally graded O-pattern (FGO); (c) uniform distribution (UD)

In Figure 5, the variation of DBLs versus α_0 and μ is presented for different PDTs and $L/h = 5$. It is seen that, increasing α_0 leads to lower DBL values.

In Figure 6, the variation of DBLs versus L/h with and without porosity, for $\alpha_0 = 0.5$ and $\mu = 2\text{nm}^2$ is displayed. Accordingly, as L/h increased the DBLs have decreased as expected. The FGX distribution exhibited the highest DBLs, whereas the FGO type yields the lowest values for FGP nanobeam. This is attributed to the fact that the FGX nanobeam has lower porosity density near the beam surfaces.

5.3 Machine Learning Analysis

This section, present the results obtained by the proposed ML framework based on parametric studies to elucidate the effect of nonlocal parameter, L/h , PDT and α_0 on DBLs of the FGP nanobeams is addressed in this section.

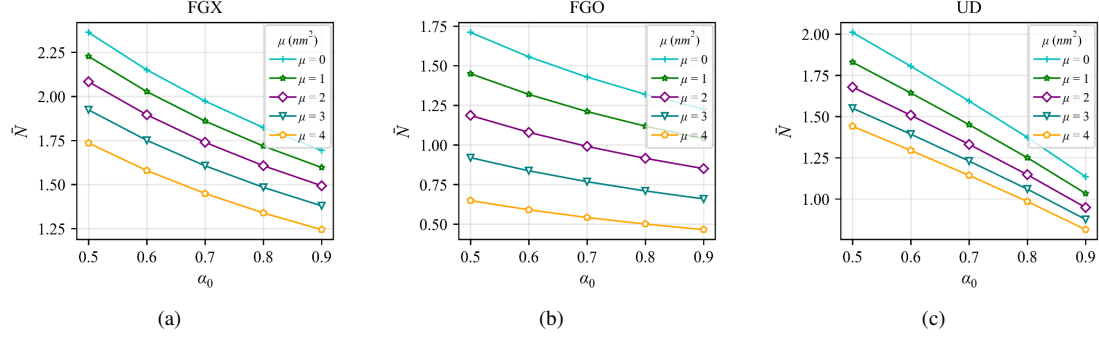


Figure 5. The variation of dimensionless buckling loads (DBLs) of functionally graded porous (FGP) nanobeam versus α_0 : (a) functionally graded X-pattern (FGX); (b) functionally graded O-pattern (FGO); (c) uniform distribution (UD)

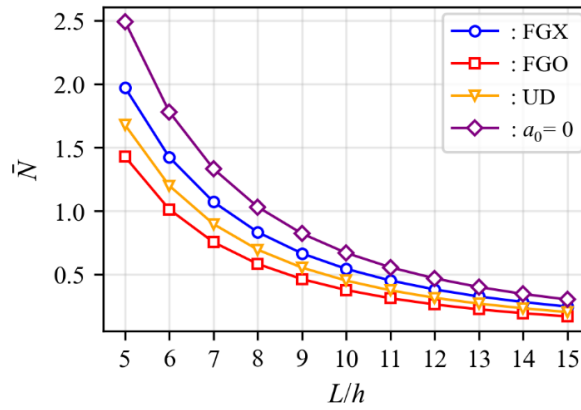
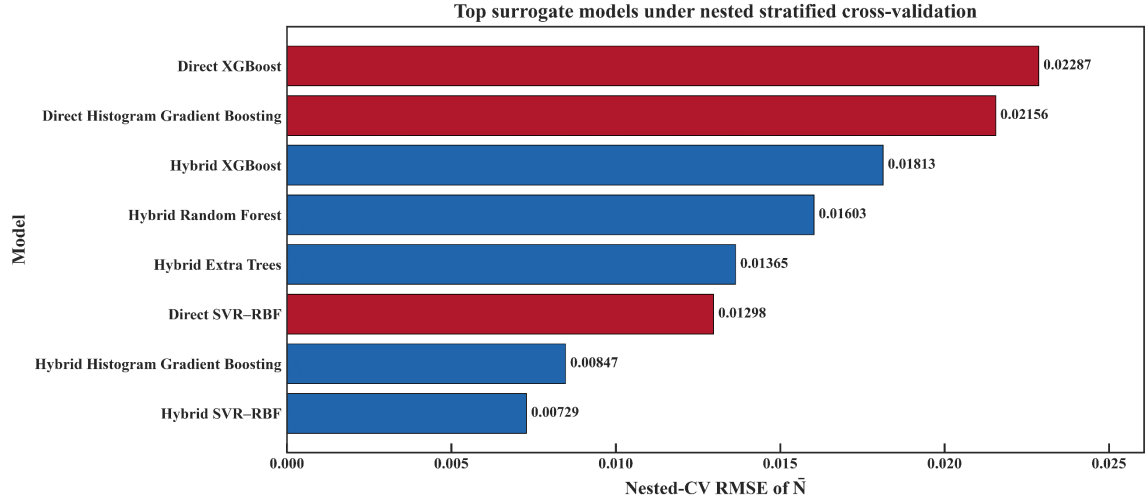


Figure 6. The variation of dimensionless buckling loads (DBLs) of functionally graded porous (FGP) nanobeam versus L/h for different porosity distribution types (PDTs)

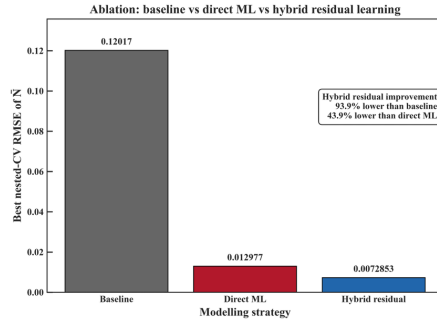
Note: FGX — functionally graded X-pattern; FGO — functionally graded O-pattern; UD — uniform distribution.

Figure 7, represent the evaluation of the proposed ML framework. As seen in Figure 7a the best models are listed and evaluated based on RMSE (when RMSE value is closer to zero, it reflects better prediction performance). As seen, spurious performance was recorded by hybrid performance SVR-RBF with RMSE = 0.00729. In overall, hybrid models revealed better performance. To clarify the enhancement which hybrid model deliver, Figure 7b demonstrates the hybrid ablation. It is illustrated that the RMSE value dropped to 0.01298 when direct ML was used, from 0.1017 obtained by the analytical baseline separately. While a further remarkable reduction was accomplished as 0.00729, when hybrid residual learning was used. Indicating enhancement in baseline and direct ML as 93.9% and 43.9%, respectively. Figure 7c represent the parity plot for the best model performance (hybrid SVR-RPF). The values of the three implemented error metrics are recorded as $R^2 = 0.999900$, MAE = 0.00309 and RMSE = 0.00729. Like RMSE, when MEA is closer to absolute zero, better prediction performance is indicated, while R^2 approaching 1 indicates better prediction performance. From the trend of scatter points in following the ideal line closely at the entire range of $\bar{N}$, and for all values of L/h it can be proven that the SVR-RBF was not only numerically the best candidate, it also success to capture the complex physics and mechanics across the entire domain. However, the reason of generating parity plot for L/h parameter not any other parameter was due to revealing its significant importance on $\bar{N}$, compared with other parameters based on permutation importance analysis and SHAP analysis. Figure 7d shows the residual plots, as seen in overall at all L/h values the minimum error was recorded by UD PDT, while increasing the complexity of porosity distribution cause to slightly increase the RMSE reflecting slightly lower performance. Yet, in overall the RMSE values for all PDTs are significantly low and accepted under this hybrid ML high validity model using nest CV and OOF. However, at $L/h = 5$ at all PDTs, indicated to exhibiting the greatest buckling loads and steeper performance gradients.

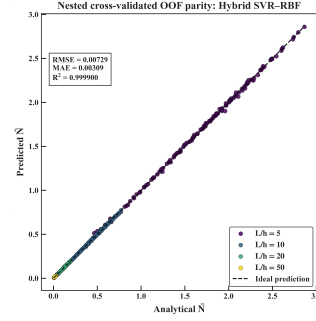
Figure 8 illustrates the uncertainty and model validation results. As seen in Figure 8a, when the training data increased, the RMSE has reduced from ≈ 0.022 to 0.005, indicating superior model performance with high fidelity. Besides, the uncertainty band shows to become thinner, reflecting higher stability. However, the training error preserved its small value through the entire processes. It is seen that there is no variation after training samples



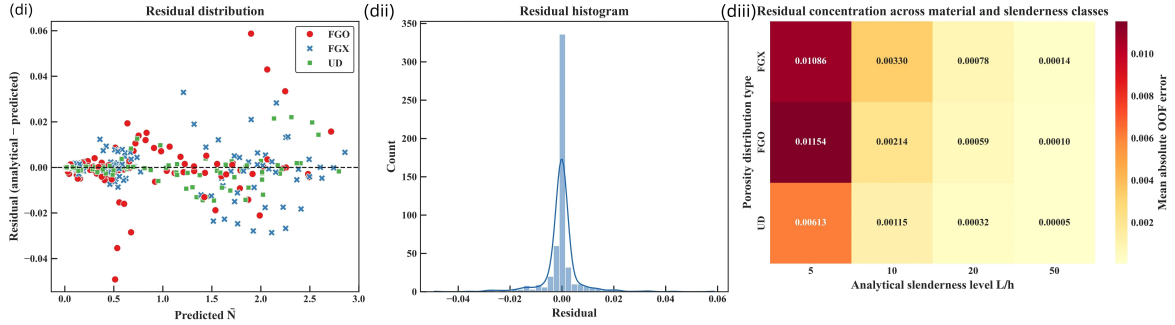
(a)



(b)



(c)



(d)

Figure 7. Evaluation of machine learning (ML) framework: (a) best surrogate models' performance under nested stratified cross-validation (CV); (b) hybrid modeling ablation baseline vs direct ML vs hybrid residual learning; (c) nested CV out-of-fold (OOF) parity plot for the best model (support vector regression-radial basis function, SVR-RBF); (d) residual distribution, histogram, and concentration across porosity distribution type (PDT) and L/h values

Note: RMSE — root mean square error; XGBoost — eXtreme Gradient Boosting; FGX — functionally graded X-pattern; FGO — functionally graded O-pattern; UD — uniform distribution.

greater than ≈ 360 . By considering the Nested CV for RMSE stability illustrated in Figure 8b, the repeated fixed-model CV and nested CV both recorders very close RMSE values as 0.00501 and 0.00525, respectively. Besides, the error of most folds seems to be ≈ 0.01 , yet few numbers of higher-error folds are seen. However, it is confirmed from the repeated analysis that the model accuracy does not depend on one favored data partition. In generally efficient stability revealed to be accomplished through the entire range of alternative validation partitions. Figure 8c prove that across the implemented hybrid models there was no noticeable desecrations of the analytical trends for all explored parameters. In contrast, the opposite scenario is seen by various direct models. This illustration confirmed that the

hybrid residual learning cause enhancements in mechanical readability and numerical accuracy. The uncertainty interval shown in Figure 8d, shows perfect agreement between prediction and analytical solution through the entire $\bar{N}$ range. A pragmatic coverage of 97.04% achieved by the 95% conformal intervals. It is worth to mention that, these intervals only quantified in the current predictive domain and does not validate any untested parameter.

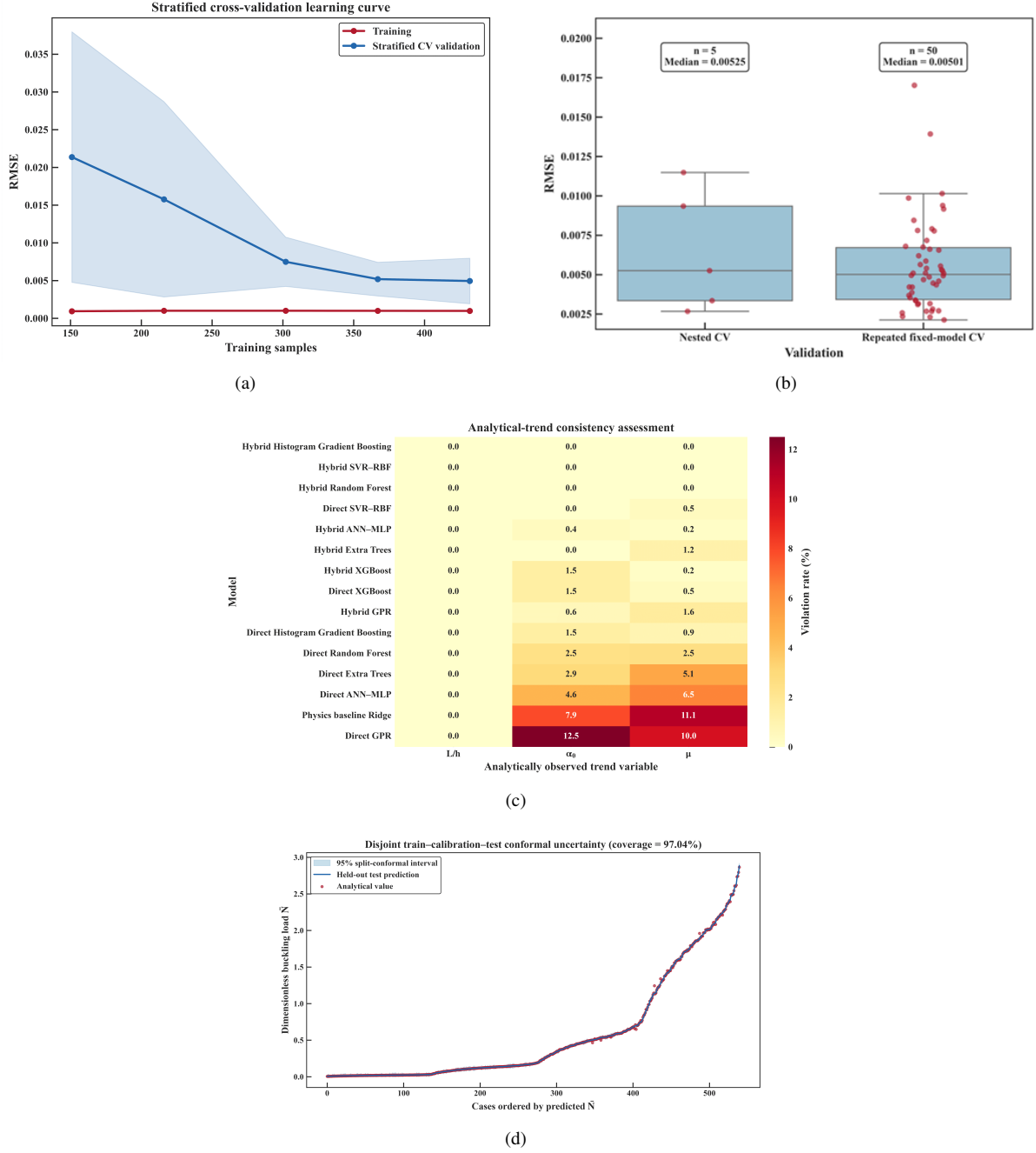


Figure 8. Uncertainty and validation of machine learning (ML): (a) stratified cross-validation (CV) learning curve; (b) root mean square error (RMSE) stability across validation folds; (c) analytical-trend consistency assessment and (d) disjoint train-calibration-test conformal uncertainty (coverage = 97.04%)

Note: SVR-RBF — support vector regression-radial basis function; ANN-MLP — artificial neural network - multi-layer perceptron; XGBoost — eXtreme Gradient Boosting; GPR — gaussian process regression.

The SHAP analysis was conducted for the best model performance only, therefore Figure 9 demonstrate the SHAP analysis implementations and how each parameter affects the final predicted $\bar{N}$. Figure 9a, shows the importance of each parameter, as seen the most important parameter is L/h while the lowest obtained to be μ . The dominant L/h indicates the robust separation of buckling load across the different four values of L/h . The secondary contribution revealed to be accomplished by α_0 . The importance results by SHAP were exactly same as permutation importance

results, signifying the reliability of interpretation. Figure 9b combined SHAP bee swarm analysis with categorizing PDT. As seen a high positive SHAP contribution accomplished by low L/h , while higher values decreased the predicted buckling load. On the other hand, a shift to negative region caused by increased α_0 and μ . The blue, red and green colors indicate the FGX, FGO and UD, respectively. Figure 9c demonstrate the SHAP interaction maps. The influence of α_0 became significantly negative with increasing the value of α_0 , which also depend on L/h . However, larger variation seems to be caused by low L/h , indicating stringer coupling or interaction between geometry-porosity at high loads area. As another evidence of the ability of the proposed model, in capturing the coupling effects between parameters rather than one single effect per time, the contribution of μ with respect to α_0 was also explored.

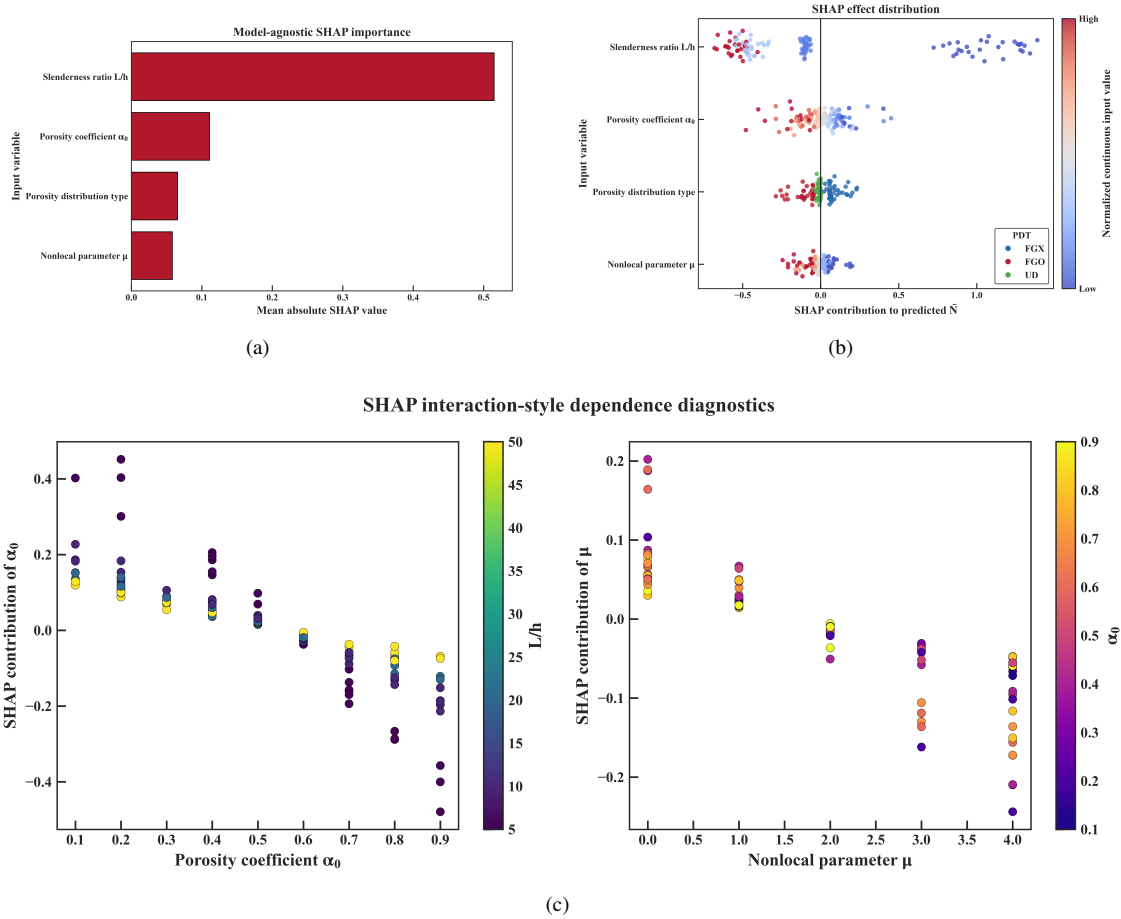
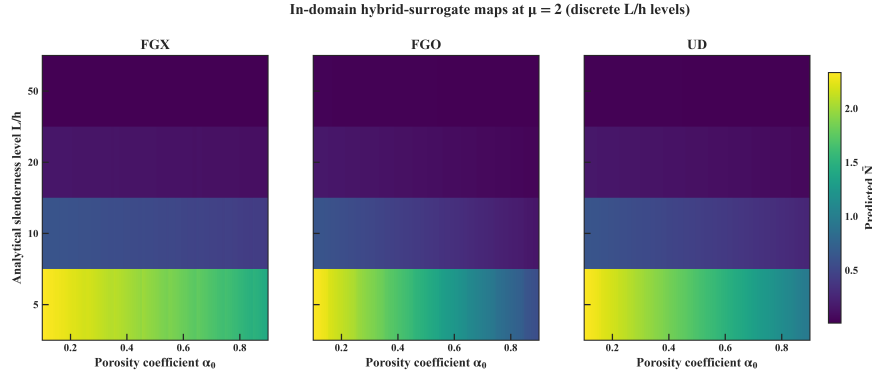


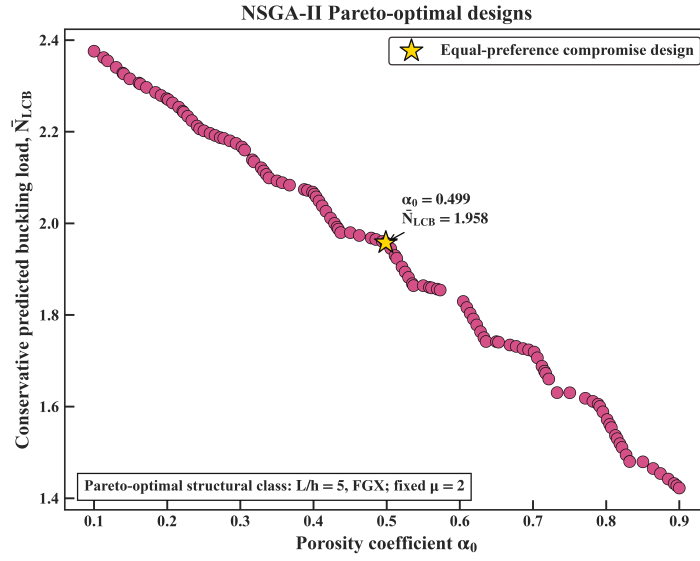
Figure 9. SHapley Additive exPlanations (SHAP) analysis: (a) SHAP importance; (b) SHAP effect distribution and (c) SHAP interaction-style dependence diagnostics

Note: PDT — porosity distribution type; FGX — functionally graded X-pattern; FGO — functionally graded O-pattern; UD — uniform distribution.

The responses surface illustrated in Figure 10a. The discrete horizontal classes reflect the L/h levels without integrating any extrapolation tool. However, it is seen that increasing L/h and α_0 , caused a drop in $\bar{N}$. At all PDT classes, the largest resistance to buckling, demonstrate to occur at low α_0 and at $L/h = 5$. In general, structures with FGX shows to have greater predicted resistance compared to FGO, mainly, at large α_0 . Meanwhile, Figure 10b demonstrates the NSGA-II Pareto front, reflecting the correlation between increasing α_0 , while preserving good buckling resistance. By considering FGX FGP nanobeam, while maintaining L/h as 5 and μ as 2, and considering α_0 as variable, as proven earlier in analytical solution the resistance to bulking reduced with increasing α_0 . However, with exploring the optimal scenario, the equal-preference compromise design seems to occur at $\alpha_0 = 0.499$ associated with average normalized local critical buckling load, $\bar{N}_{LCB} = 1.958$, at which a balance between buckling resistance and α_0 can be accomplished.



(a)



(b)

Figure 10. (a) Surface maps responses and (b) Nondominated Sorting Genetic Algorithm II (NSGA-II) Pareto-optimal designs

Note: FGX — functionally graded X-pattern; FGO — functionally graded O-pattern; UD — uniform distribution.

6 Conclusions

This study developed a hybrid analytical–ML framework for predicting and interpreting the buckling behaviour of FGP nanobeams. The analytical formulation combined exponential shear deformation theory with Eringen’s nonlocal elasticity theory. The governing equations were derived from the minimum total potential energy principle and solved by Navier’s method for simply supported boundary conditions. Three porosity distributions based on the Gibson–Ashby model were considered, and the analytical formulation was verified against published benchmark results. The resulting dataset of 540 configurations was then used to construct a residual-learning surrogate that combined a regularised Ridge baseline with nonlinear ML models. Nested CV, conformal prediction, SHAP analysis, and NSGA-II optimisation were incorporated to evaluate prediction accuracy, quantify uncertainty, interpret parameter effects, and support design exploration.

The principal findings are summarised as follows:

- The nonlocal formulation predicted lower DBLs than the corresponding local limit. The DBL decreased consistently as the nonlocal parameter increased, confirming the importance of small-scale effects in the stability analysis of FGP nanobeams.
- Increasing the porosity coefficient reduced the DBL, although the magnitude of this reduction depended on the porosity distribution. Among the distributions examined, FGX produced the highest buckling resistance, FGO produced the lowest, and UD showed intermediate behaviour.
- The DBL decreased with increasing slenderness ratio. The interaction between the slenderness ratio and porosity coefficient was particularly pronounced in configurations associated with relatively high buckling loads.

- The exponential shear deformation formulation satisfied the traction-free conditions at the upper and lower surfaces without requiring a shear correction factor. It therefore provided a consistent representation of transverse shear deformation within the analytical framework adopted in this study.

- The hybrid SVR-RBF surrogate produced the best predictive performance among the examined models, with an RMSE of 0.00729, an MAE of 0.00309, and an R^2 of 0.999900. It also outperformed both the regularised baseline and the corresponding direct ML formulation.

- The learning-curve, repeated-validation, and leave-one-level-out analyses showed that the surrogate retained the principal mechanical trends of the analytical model across the investigated parameter domain. The 95% conformal prediction intervals achieved an empirical coverage of 97.04%, indicating reliable uncertainty estimates for configurations within this domain.

- The SHAP analysis identified the slenderness ratio as the most influential input, followed by the porosity coefficient, while the nonlocal parameter had the lowest overall contribution within the selected ranges. These interpretations were consistent with the trends obtained from the analytical parametric analysis.

- Coupling the validated surrogate with NSGA-II produced a Pareto front describing the trade-off between porosity and buckling resistance. For the FGX distribution at $L/h = 5$ and $\mu = 2$, the equal-preference solution was obtained at $\alpha_0 = 0.499$, with a predicted DBL of 1.958.

The proposed framework shows how analytical mechanics, residual learning, uncertainty quantification, interpretable modelling, and multi-objective optimization can be coordinated within a single design procedure. It provides a computationally efficient means of evaluating FGP nanobeam configurations within the investigated parameter space while retaining the mechanical trends of the underlying analytical formulation.

The present study is limited to the linear static buckling of simply supported FGP nanobeams represented by three prescribed porosity distributions. In addition, the surrogate was trained entirely on analytically generated data and was not independently assessed against experimental measurements or high-fidelity numerical simulations. Future work should therefore consider other boundary conditions, grading patterns, loading conditions, nonlocal formulations, and beam theories. Extending the framework to bending, vibration, thermal buckling, and post-buckling responses, together with validation using finite-element or experimental data, would provide a stronger basis for its application to practical nano structural design.

Author Contributions

Conceptualization, M.A.; methodology, M.A.; software, M.A.; validation, M.A.; formal analysis, B.İ. and M.A.; investigation, B.İ. and M.A.; writing—original draft preparation, B.İ.; writing—review and editing, M.A.; visualization, B.İ.; supervision, M.A.; project administration, M.A. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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